

RESTAURANT KITCHEN ORDER SCHEDULER

Smart Kitchen Order Scheduling Using DAA

PRIORITY QUEUE • GREEDY RULE • DEADLINES

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ORDERS

DEADLINES

PRIORITY

A clearer sequence for every
ticket.

THE CHALLENGE IN A BUSY KITCHEN

When several orders arrive together, what should the kitchen prepare first?

01 Multiple orders

Several tickets can arrive at the same time.

02 Different prep times

Every dish occupies the kitchen for a different duration.

03 Different urgency

Some orders need attention before others.

04 Promised deadlines

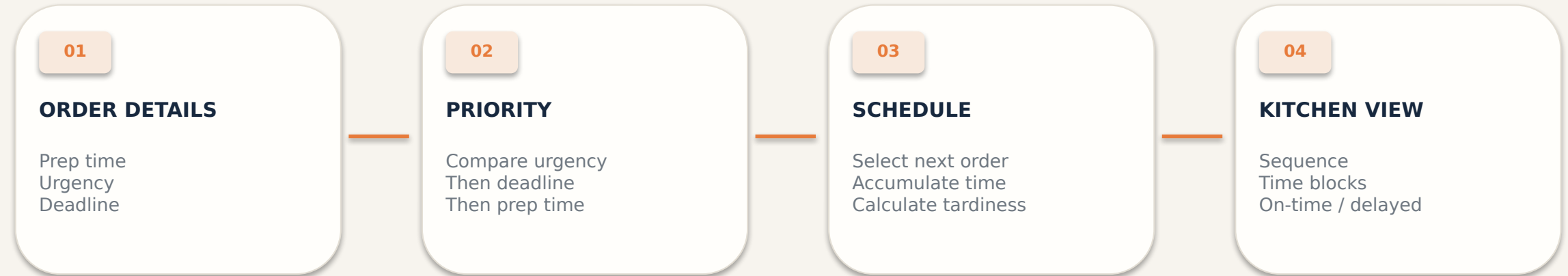
Late completion can affect pickup and delivery.



POOR SEQUENCING → LATE ORDERS → CUSTOMER FRICTION

FROM ORDER DETAILS TO A KITCHEN SCHEDULE

One visible workflow turns order attributes into an explainable sequence.

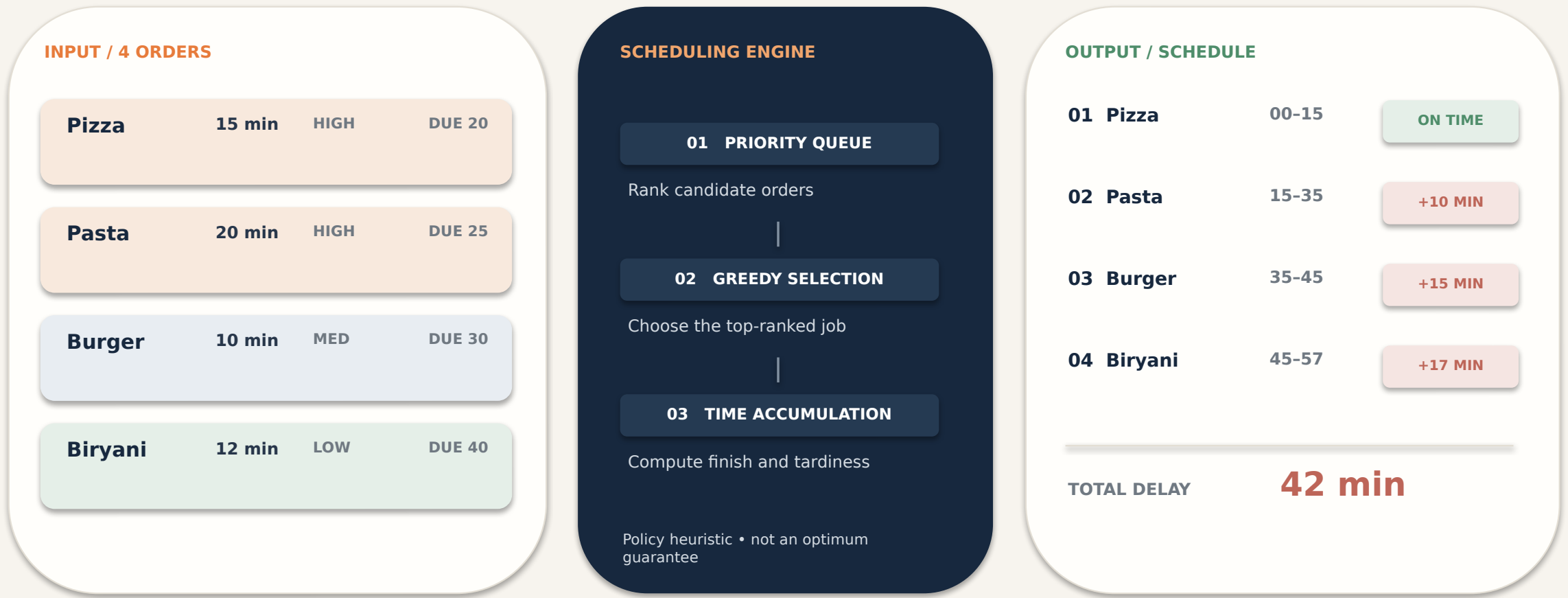


DESIGN GOAL

Make prioritization explicit and show where the schedule may miss a deadline.

INPUT → ALGORITHM → OUTPUT

A single-workstation example using the order data from the project brief.



THE LOGIC BEHIND THE SCHEDULE

Three familiar DAA ideas work together as an explainable scheduling heuristic.

01

PRIORITY QUEUE

Keeps the next candidate easy to identify.

Key: urgency → deadline → prep

Heap operations: $O(\log n)$

02

GREEDY RULE

Chooses the highest-ranked available order at each step.

Simple, deterministic policy

Does not guarantee global optimum

03

JOB SCHEDULING

Maps selected orders to consecutive time intervals.

Finish = start + prep time

Tardiness = $\max(0, \text{finish} - \text{due})$

The purpose is transparent prioritization—not a claim of mathematically minimum delay.

HOW DOES THE SYSTEM CHOOSE THE NEXT ORDER?

A fixed rule makes each selection explainable and repeatable.

DECISION HIERARCHY

1 HIGHER URGENCY

High → Medium → Low

2 EARLIER DEADLINE

Compare due times

3 SHORTER PREP

Break remaining ties

Final tie: stable order ID / entry order

WORKED EXAMPLE

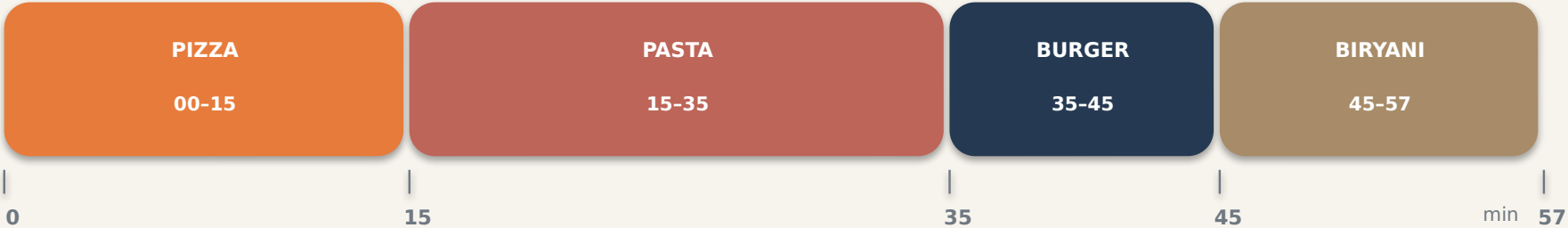
ID	DISH	PREP	URGENCY	DEADLINE
01	Pizza	15 min	HIGH	20 min
02	Pasta	20 min	HIGH	25 min
03	Burger	10 min	MEDIUM	30 min
04	Biryani	12 min	LOW	40 min

FIRST PICK: 01 Pizza • High urgency, earliest deadline among High

FROM ORDERS TO A KITCHEN TIMELINE

Completion is compared with each order’s deadline; delays are calculated in minutes.

SINGLE WORKSTATION / TIME FROM SCHEDULING START



DEADLINES

PIZZA	20 min due	PASTA	25 min due
BURGER	30 min due	BIRYANI	40 min due

Calculated from the example data; this is not a measured software benchmark.

SUMMARY

57 min

total prep

1 ON TIME

3 DELAYED

42 min

total tardiness

A MORE ORGANIZED WAY TO RUN THE TICKET QUEUE

The scheduler makes the trade-offs visible instead of hiding them in a final sequence.

01 DEADLINE AWARE

Surfaces which orders are likely to finish late.

02 URGENT WORK VISIBLE

Allows staff to express importance explicitly.

03 CLEAR SEQUENCE

Gives the kitchen one shared preparation order.

04 EXPLAINABLE CHOICES

Shows the factors behind each selection.

05 BETTER TIME VIEW

Makes cumulative workload easier to inspect.

06 RECALCULATABLE

Can refresh the unstarted plan when tickets change.

BETTER SCHEDULING → BETTER KITCHEN COORDINATION → BETTER SERVICE

THE SAME SCHEDULING IDEA TRAVELS

Urgency, processing time, deadlines, and limited capacity appear in many workflows.

01

RESTAURANTS

Prepare food orders by urgency, prep time, and promised pickup.

FOOD SERVICE

02

HOSPITALS

Prioritize tasks by urgency while accounting for available resources.

CARE WORKFLOWS

03

E-COMMERCE

Sequence picking, packing, and dispatch against shipment cutoffs.

FULLFILLMENT

04

MANUFACTURING

Schedule production jobs around processing duration and due dates.

PRODUCTION

Each domain needs its own constraints and priority policy; the restaurant rule should not be copied blindly.

TECHNOLOGY & SYSTEM ARCHITECTURE

A small web stack keeps the interface separate from the scheduling logic.

PROPOSED STACK

FRONTEND

HTML / CSS / JavaScript

BACKEND

Python / Flask

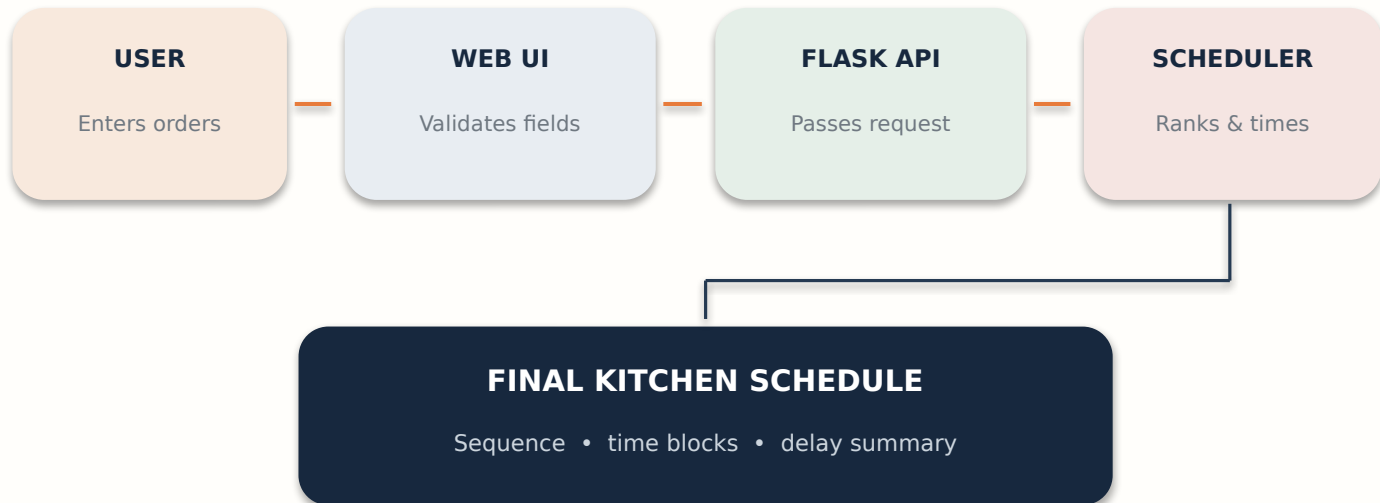
ALGORITHM

Priority queue / greedy rule

DATA STRUCTURE

Heap or sorted order list

REQUEST FLOW



Intended architecture from the supplied brief; implementation status is not verified.

FROM BASIC SCHEDULING TO SMARTER OPERATIONS

Useful extensions should reflect the real kitchen's people, stations, and order flow.

01 REAL-TIME UPDATES

Recalculate unstarted work when a new ticket arrives.

02 MULTIPLE CHEFS

Assign jobs across chefs and workstations.

03 DELIVERY STATUS

Connect kitchen progress with dispatch timing.

04 PREP-TIME ESTIMATES

Learn duration ranges from historical orders.

05 MOBILE VIEW

Let staff check the current schedule on the floor.

06 KITCHEN ANALYTICS

Track lateness, workload, and policy trade-offs.



ALGORITHMIC SCHEDULING → INTELLIGENT RESTAURANT OPERATIONS

SMARTER SCHEDULING. FASTER SERVICE.

A practical demonstration of how DAA ideas can structure real order decisions.



“From a list of orders to a smarter kitchen schedule.”

THANK YOU

DAA HACKATHON